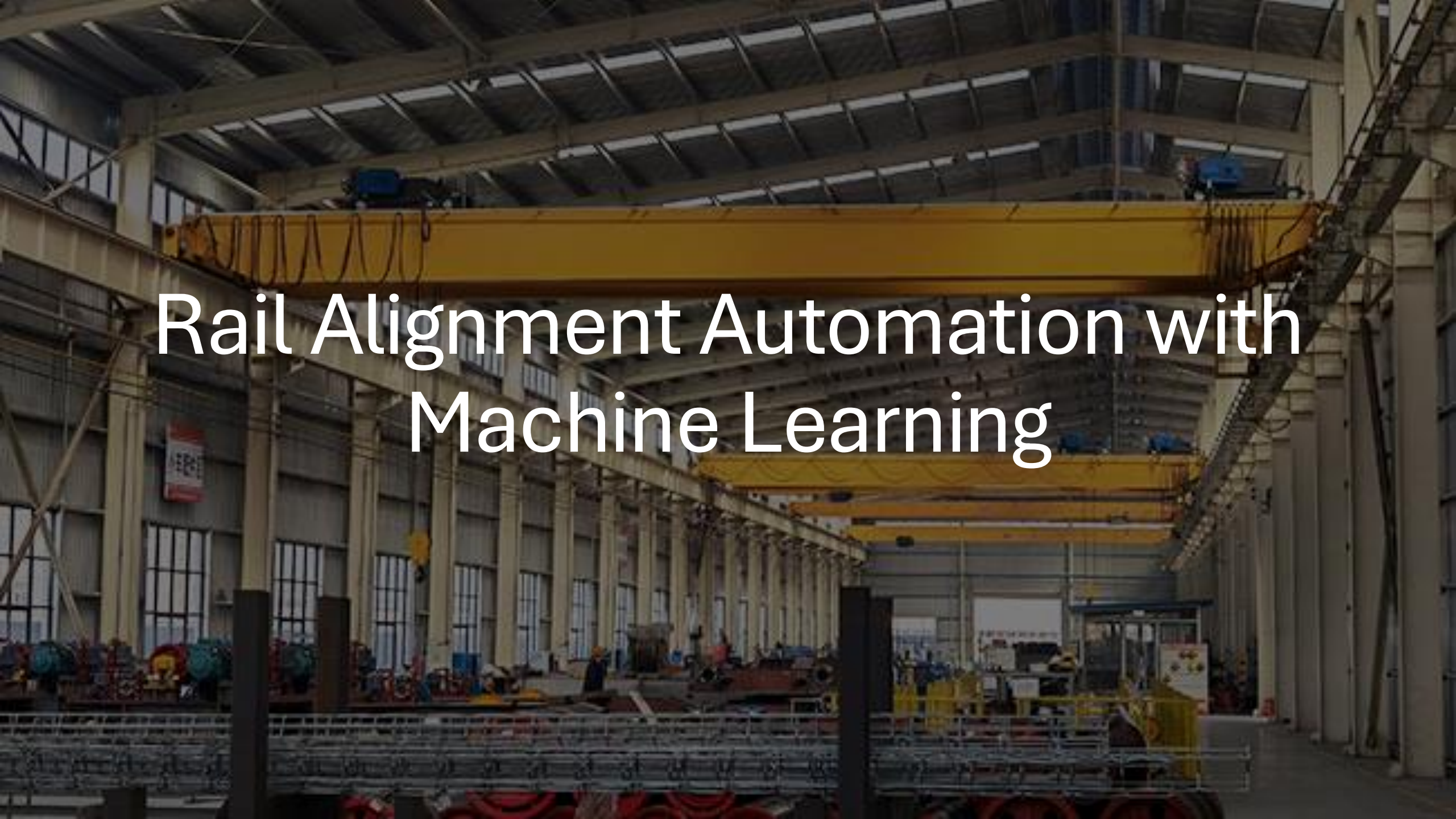


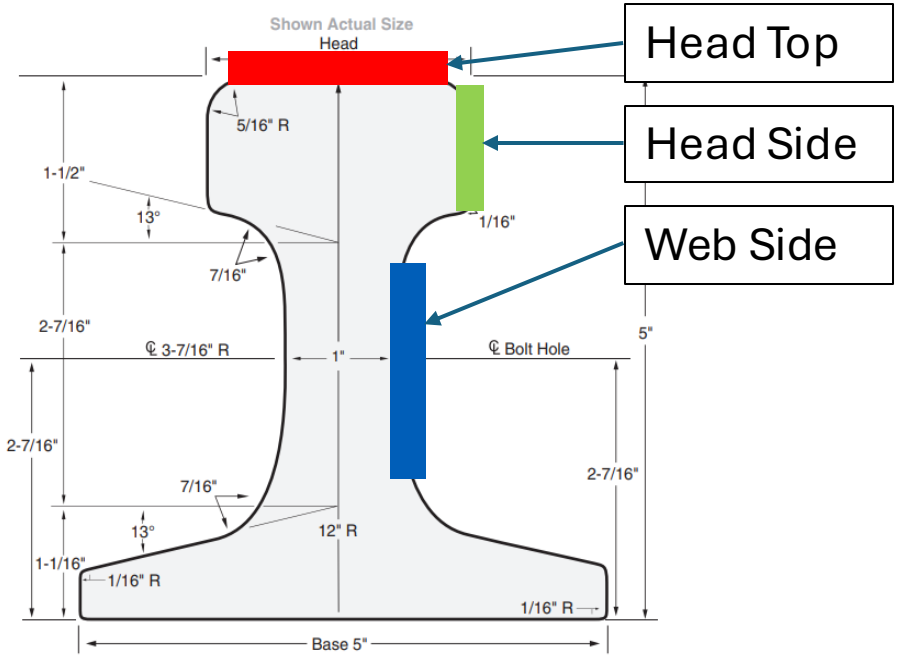
Rail Alignment Automation with Machine Learning

Machine Learning – Rail Alignment

Reason for Project

- **Problem – Misalignment of rails to crane wheels**
  - Causes Premature Wear
  - Operation Downtime

- **Goal – Use 3D scanner data for rail adjustments**
  - Classify and isolate rail points from rest of scan
  - Import rail head and rail web into data frames for analysis
  - Analyze rail with CMAA standards



Machine Learning – Rail Alignment

Project Background

- **The Why:**

To develop rail alignment adjustments for rail straightness using machine learning (ML) models and 3D survey scan data **so that** rail adjustments are automated and have reduced error

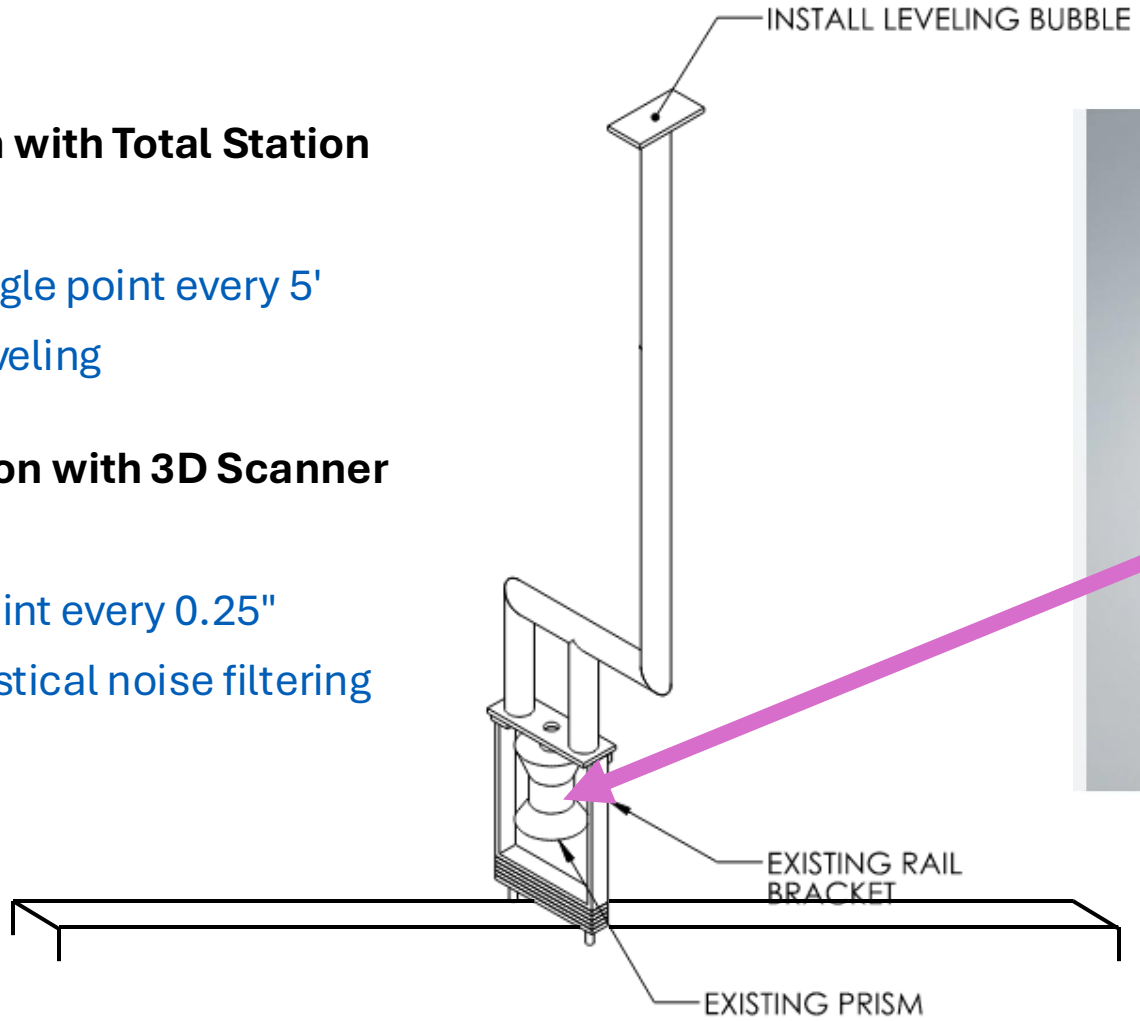
- **The How:**
  - 3D scanner (Leica BLK360) captures XYZ coordinates at each location it is places
  - Scanner uses a high accuracy laser to measure distance to object while rotating
  - Data processing with Supervised and Unsupervised ML models

BL360 Scanner

Machine Learning – Rail Alignment

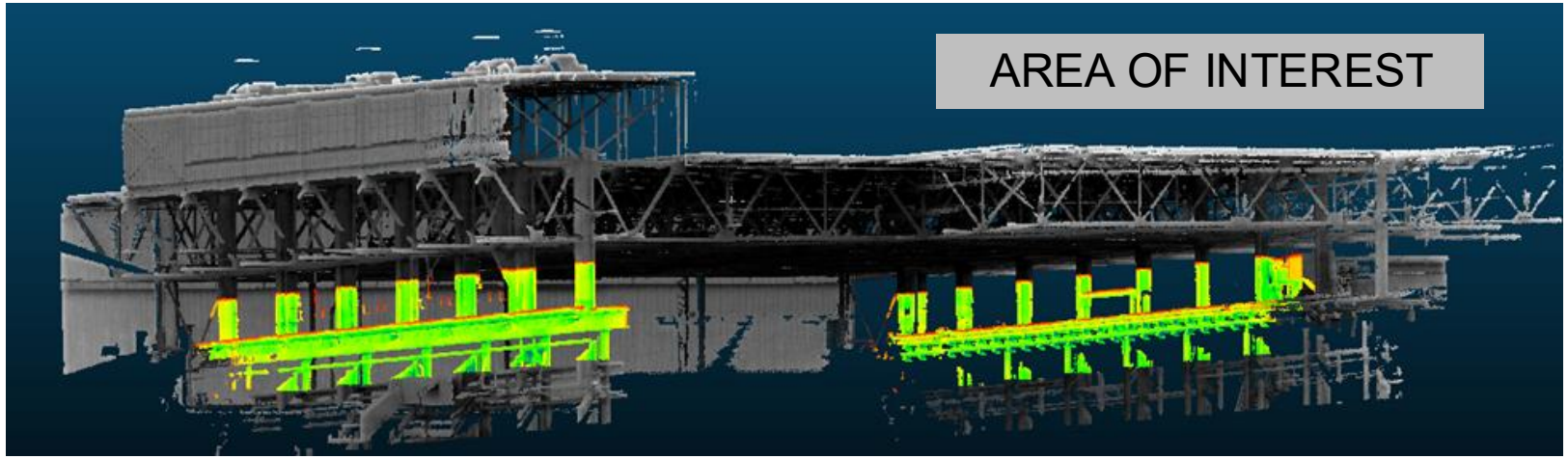
Old vs New Method

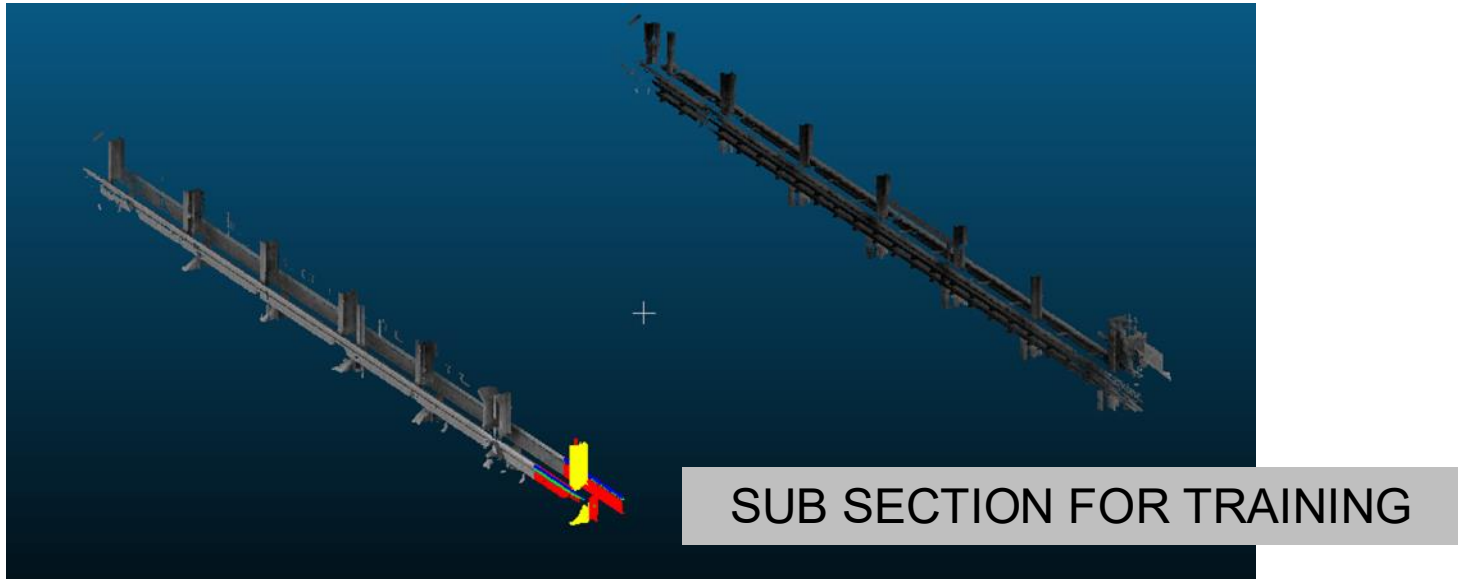
- **Current Point Collection with Total Station**
  - Scanner on Ground
  - Low Resolution - Single point every 5'
  - Human error with leveling
- **Proposed Point Collection with 3D Scanner**
  - Scanner of Crane
  - High Resolution – Point every 0.25"
  - Minimal error – Statistical noise filtering



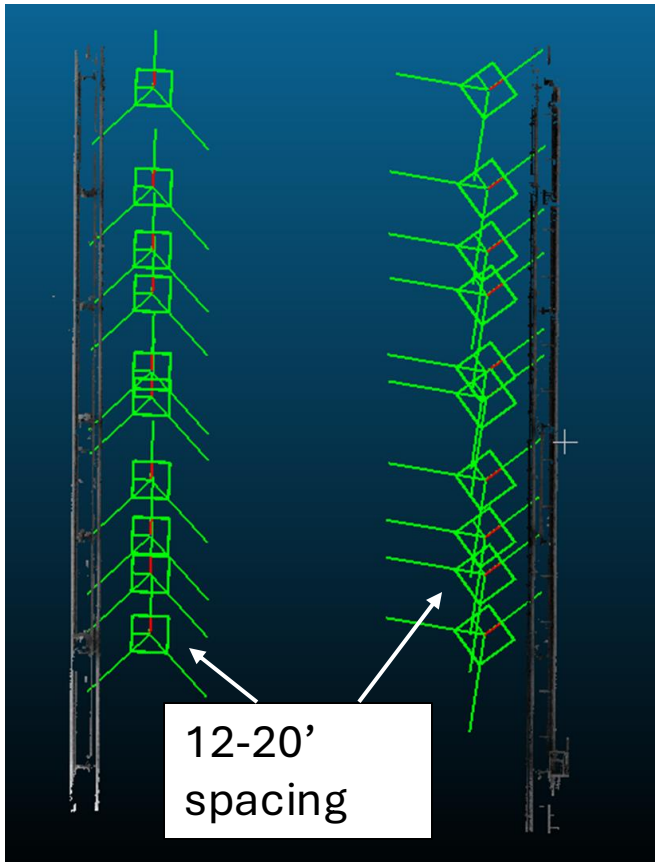
Machine Learning – Rail Alignment

Scan Data Overview





SCANNER LOCATIONS



Machine Learning – Rail Alignment

Feature Development

MATRIX MATH

Matrix of left eigenvectors (one eigenvector per row)

Matrix of right eigenvectors (one eigenvector per column)

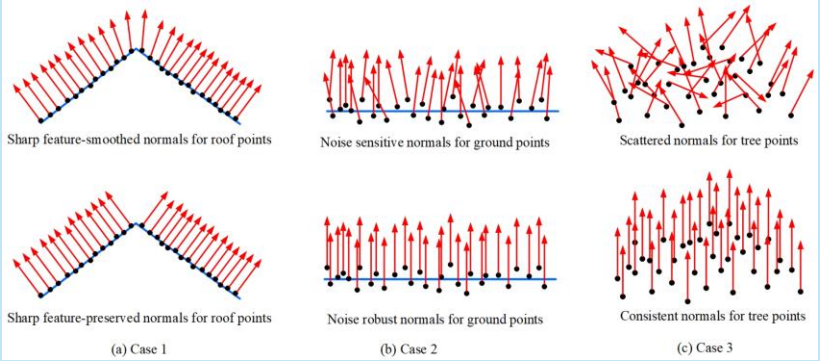
$$\mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$$

Diagonal matrix containing eigenvalues

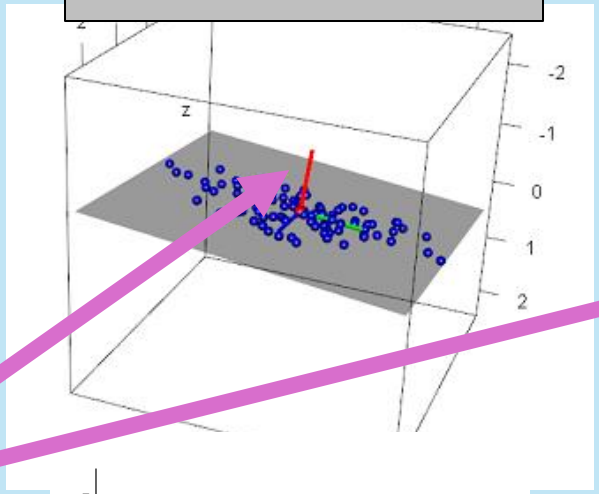
Theorem (Perron-Frobeius theorem)

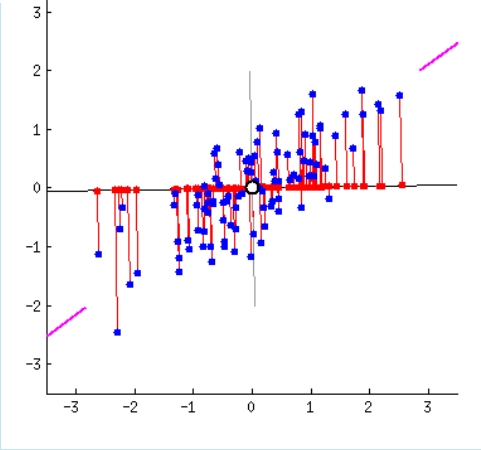
Any square matrix with all positive real elements:

- *is diagonalizable :* $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$
- *has a unique largest eigenvalue* $\lambda_1 > \lambda_2 > \lambda_3 > \cdots > \lambda_n$



LOCAL BEST FITS



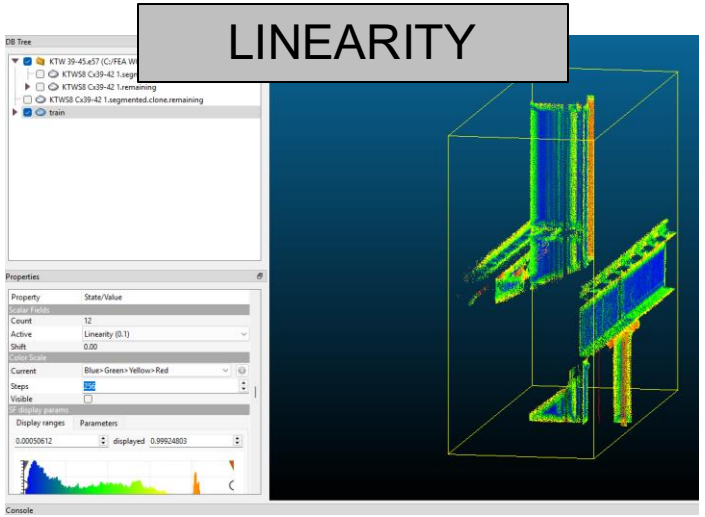


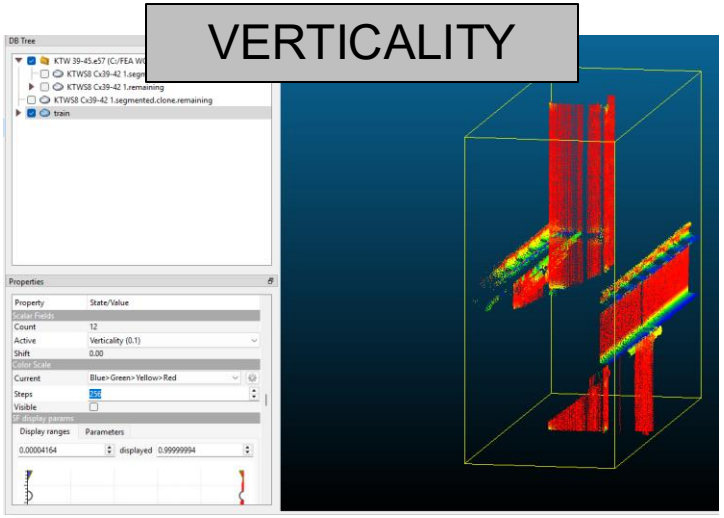
GEOMETRY FILTERS

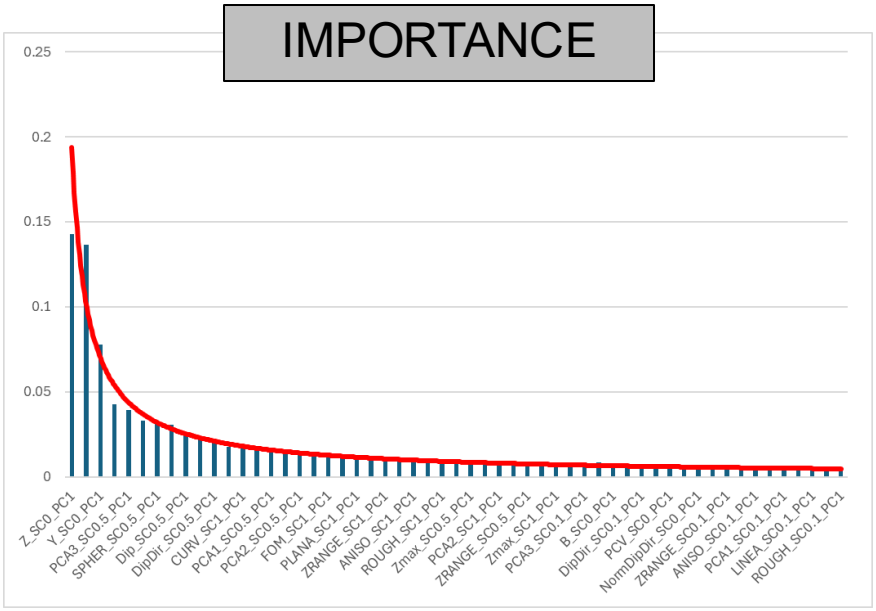
Sum	$\lambda_1 + \lambda_2 + \lambda_3$
Omnivariance	$(\lambda_1 \cdot \lambda_2 \cdot \lambda_3)^{\frac{1}{3}}$
Eigenentropy	$-\sum_{i=1}^{3} \lambda_i \cdot ln(\lambda_i)$
Anisotropy	$(\lambda_1 - \lambda_3)/\lambda_1$
Planarity	$(\lambda_2 - \lambda_3)/\lambda_1$
Linearity	$(\lambda_1 - \lambda_2)/\lambda_1$
Surface Variation	$\lambda_3/(\lambda_1 + \lambda_2 + \lambda_3)$
Sphericity	λ_3/λ_1
First Order Moment	O, see Eq. (2)
Line Feature	Q, see Eq. (3)
Orientation Feature	R, see Eq. (4)
Verticality	$1 - \lvert\langle[0\,0\,1], \mathbf{e}_3\rangle\rvert$

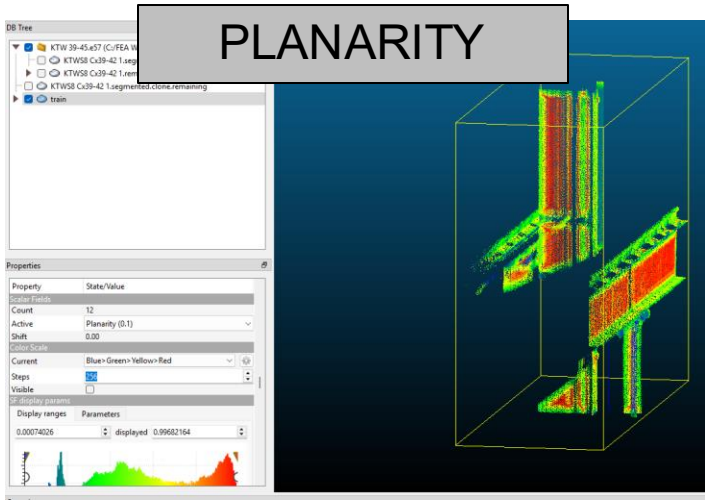
Machine Learning – Rail Alignment

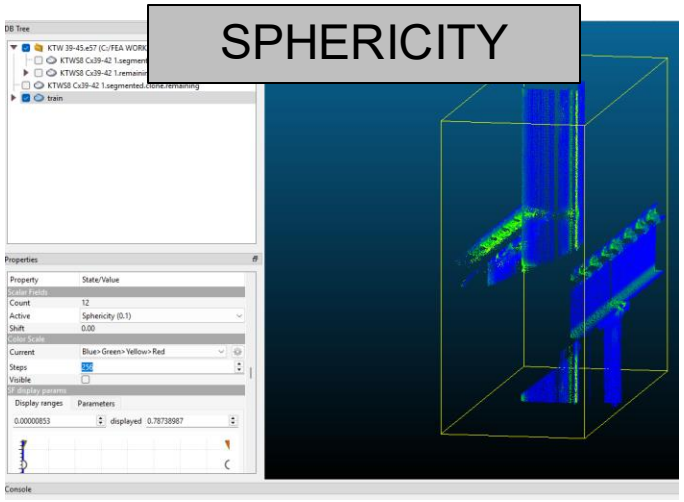
Feature Ranking

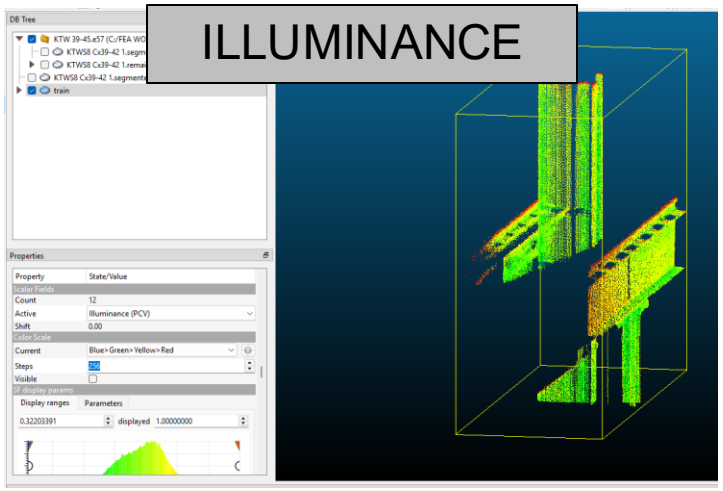






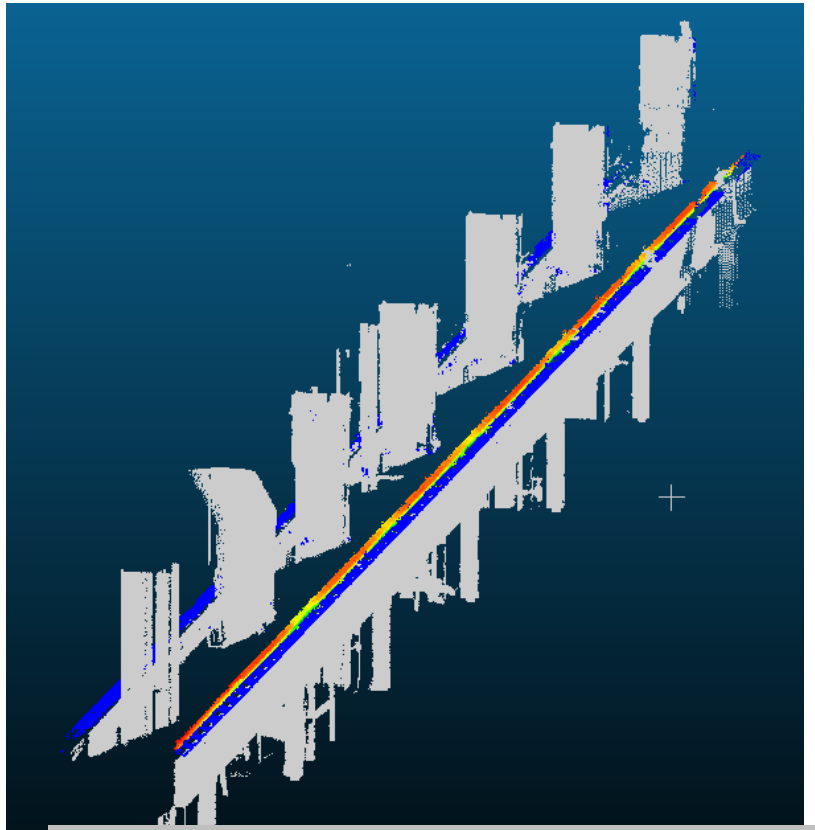






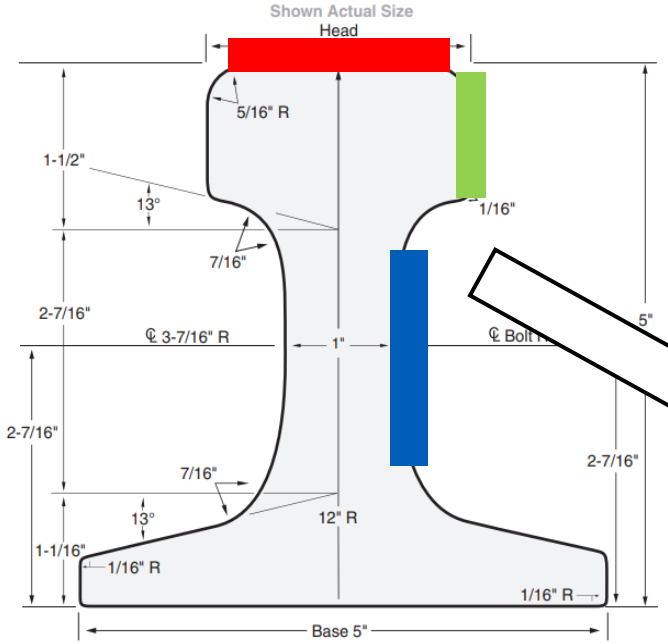
Machine Learning – Rail Alignment

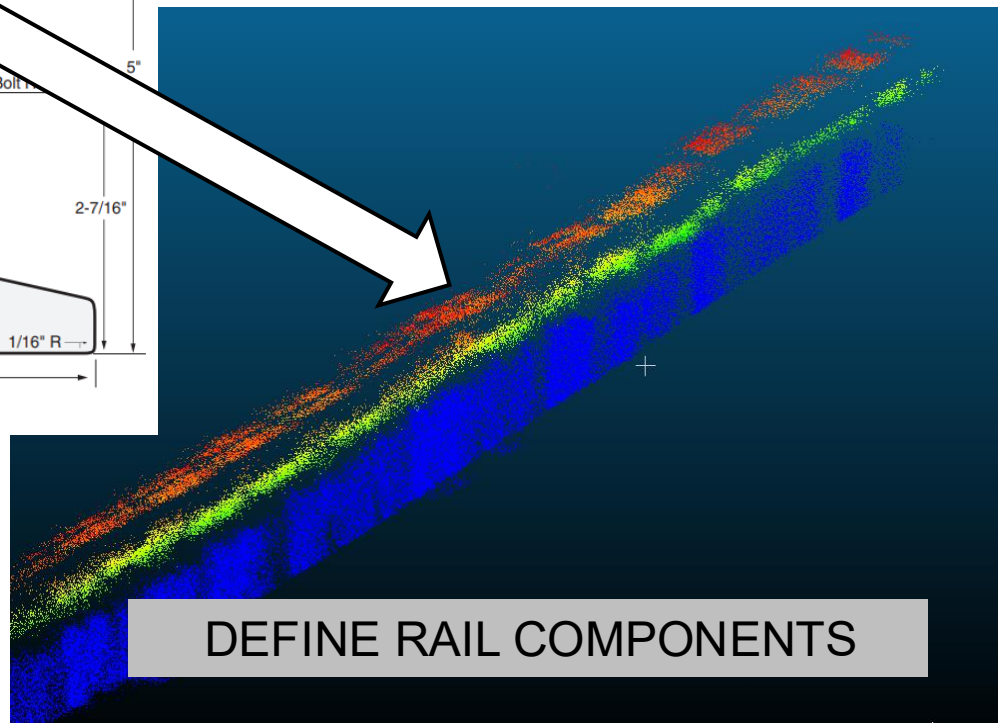
Isolation of Rail Components



TRAINED CLASSIFIER ISOLATES RAIL AND SUPPORTS

RAIL PROFILE





DEFINE RAIL COMPONENTS

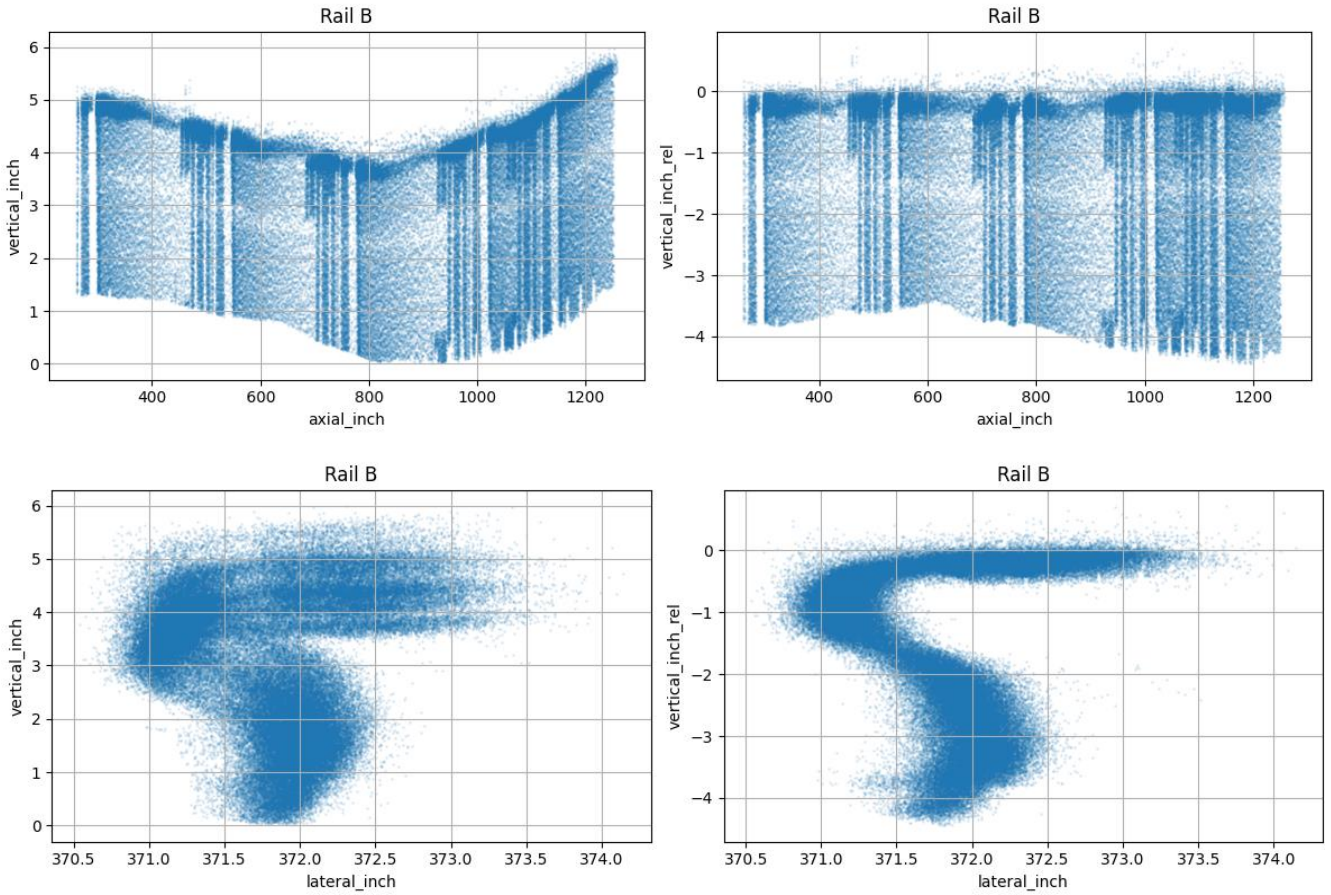
Machine Learning – Rail Alignment

Normalized Straightness

- **Improve Geometric Features with:**
  - Aligning all points to horizontal top
  - Adding synthetic points to fill gaps

Flatten Curve

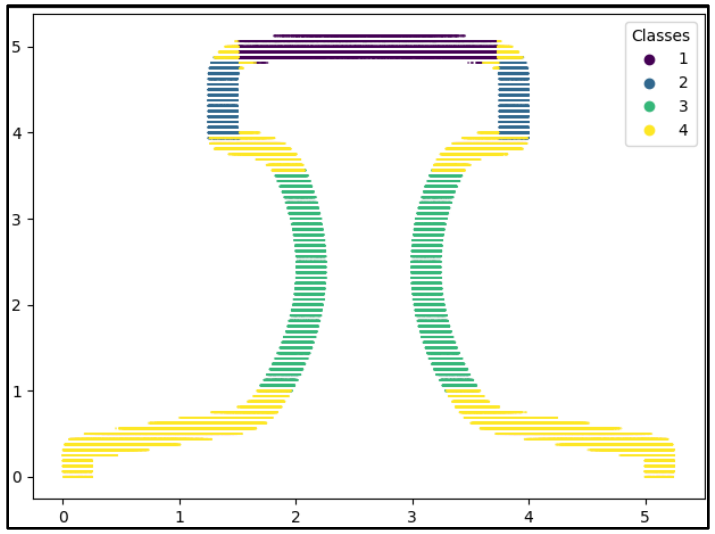
Fill Gaps



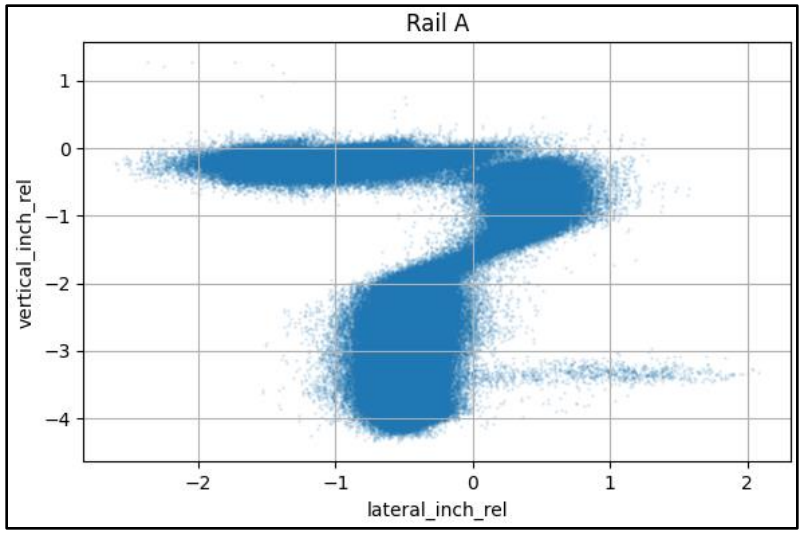
Machine Learning – Rail Alignment

Supervised Learning (Initial Isolation)

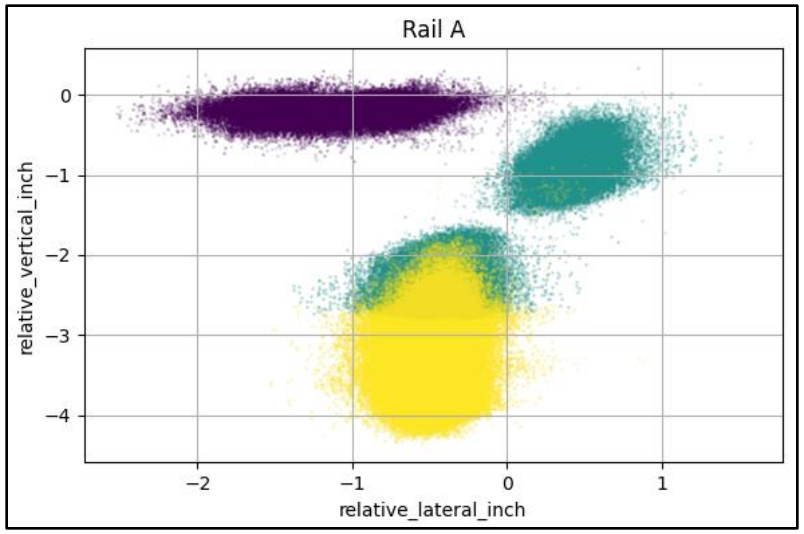
IDEAL TRAINING RAIL PROFILE



RAIL PROFILE BEFORE CLASSIFICATION



RAIL PROFILE AFTER CLASSIFICATION



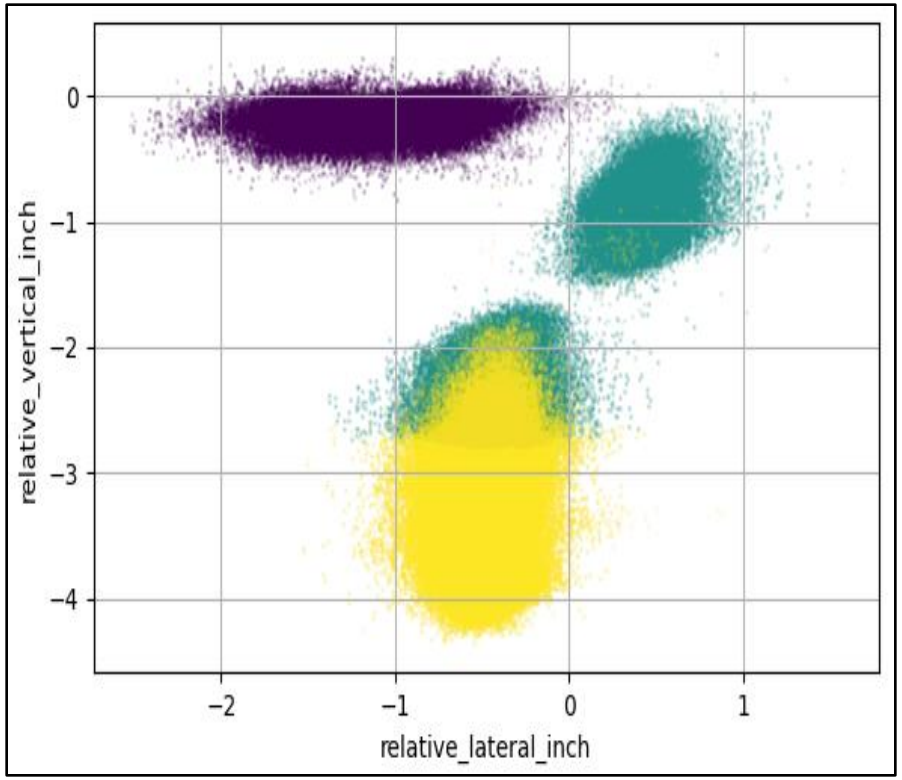
Applied Random Forest Model

Results

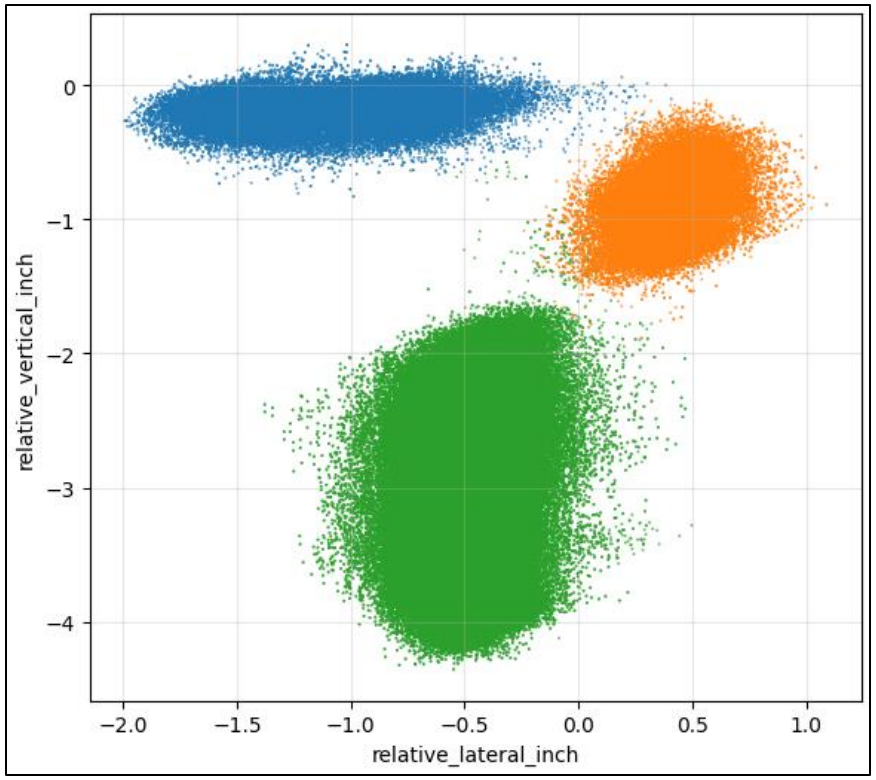
Machine Learning – Rail Alignment

Unsupervised Learning (Final Isolation)

RAIL PROFILE
BEFORE RE-CLUSTERING

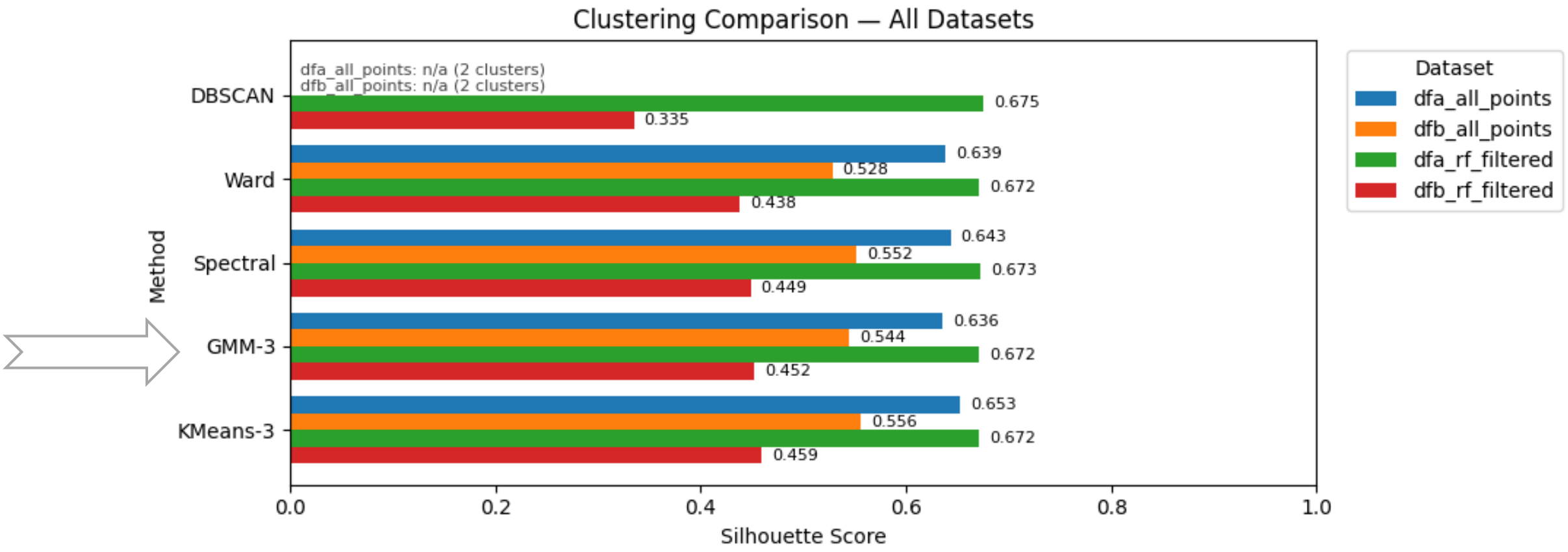


RAIL PROFILE
AFTER RE-CLUSTERING



Machine Learning – Rail Alignment

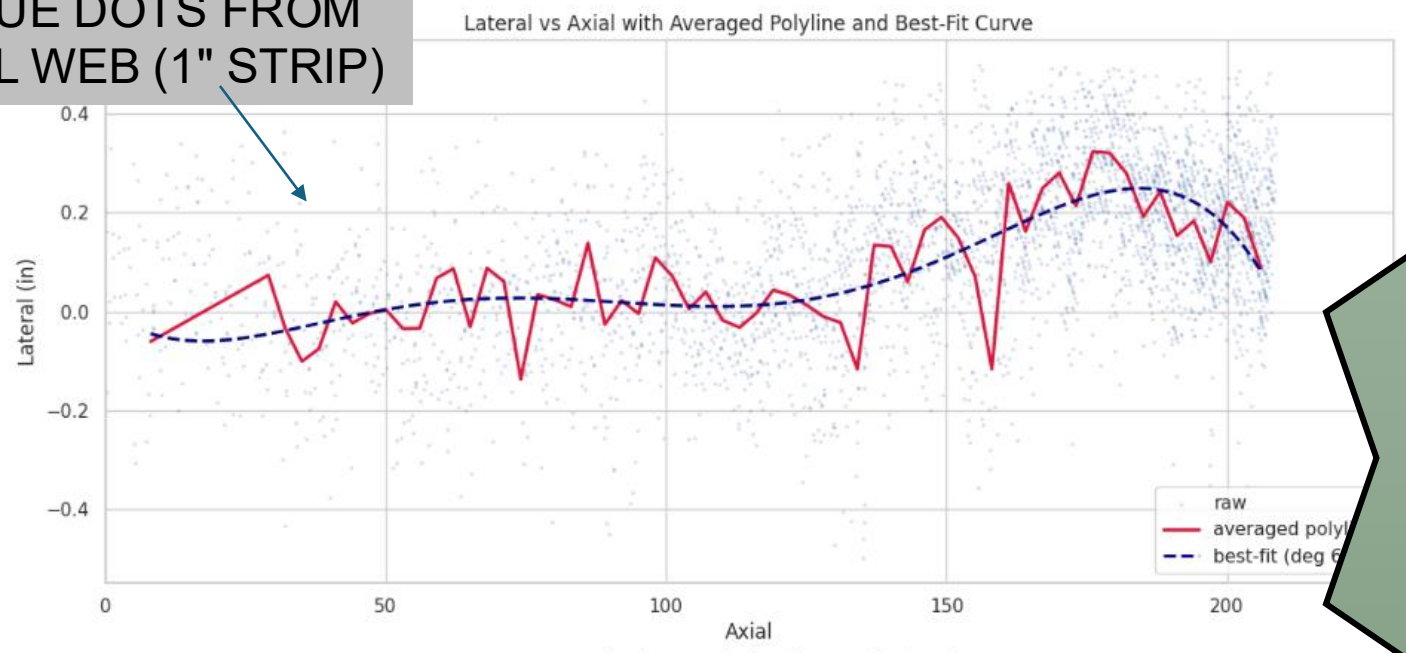
Unsupervised Learning (Model Selection)

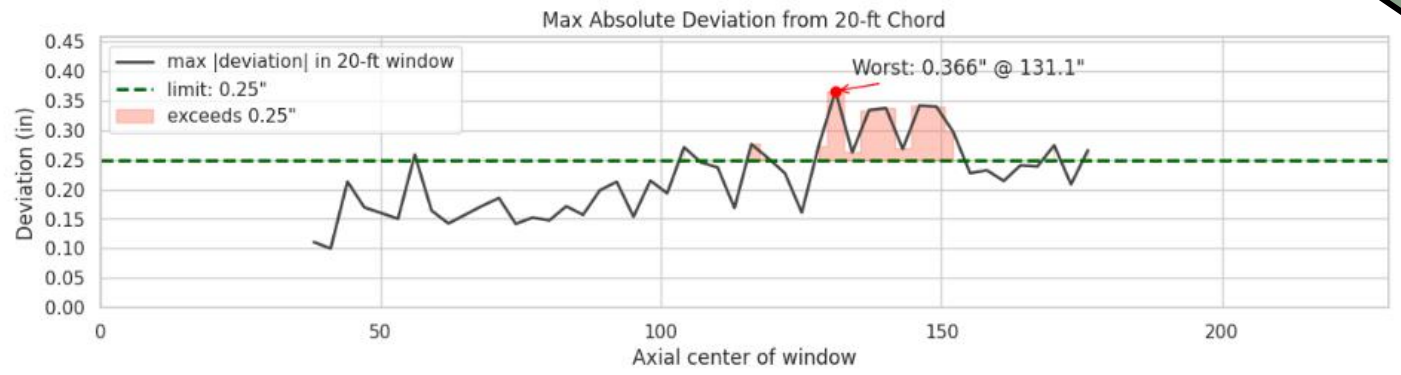


Machine Learning – Rail Alignment

Analysis Graphs Against CMAA

BLUE DOTS FROM RAIL WEB (1" STRIP)





BENEFITS

- AUTOMATION!
- IMPROVED ACCURACY!
- QUICK RESULTS!